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Determinants of customer satisfaction in chatbot use: evidence from a banking application in Turkey

Berrin Arzu Eren
Department of Accounting and Tax Applications, Vocational School, Ufuk University, Ankara, Turkey

Abstract

Purpose – This study aims to investigates customer satisfaction from the use of bank chatbots and the effect of perceived trust in chatbots and banks' reputation on customer satisfaction.

Design/methodology/approach – A survey is conducted in Turkey involving 240 customers who experienced banking transactions using a chatbot. Partial least squares structural equation modeling (PLS-SEM) is used to investigate the relationships between the variables. The data were analyzed using SPSS 21 and SmartPLS programs.

Findings – Perceived performance, perceived trust and corporate reputation significantly affect customer satisfaction with chatbot use. Customer expectations and confirmation of customer expectations have no direct impact on customer satisfaction, but customer expectations positively affect perceived performance. Customer expectations exert an indirect influence on customer satisfaction through perceived performance. Perceived performance has a positive impact on the confirmation of customer expectations, but customer expectations do not significantly impact the confirmation of customer expectations.

Research limitations/implications – This study relies on a limited number of participants. Moreover, its sample is not representative of the target population due to the convenience sampling technique. Even if the results may not be generalized to the entire population of Turkey, they reflect the reality of emerging markets with relatively high technology sensitivity and a young population.

Practical implications – The results provide new insights regarding banking service delivery channels, which may be of interest to professionals, academics, banks' top management, product development teams, design teams and customer satisfaction units.

Social implications – This study is believed to help the community make their lives easier by providing them with knowledge and awareness about chatbots.

Originality/value – This study extends expectations confirmation theory's predictions to chatbot use in banking.

Keywords Chatbot technology, Expectations confirmation theory, Perceived trust, Corporate reputation, Customer satisfaction

Paper type Research paper

Introduction

In this century, often referred to as the digital age, human beings have reached the "second machine age" using robots and artificial intelligence (AI) in their daily life and at work (Brynjolfsson and McAfee, 2014). "Big data, machine learning, deep learning, and super-intelligence have gained oil-like value in this age" (Helbing, 2019, p. 48). AI is also a crucial element of customer experience. Owing to AI, it has become possible to filter, analyze and interpret customer data that are too large for a human to process (Oracle, 2017).

The banking sector has increasingly adapted to the digital age, even though it has been essentially human-powered in the past. With the advent of new technologies, financial products and services are gathered under the umbrella of Fintech. Unlike the internet or mobile banking and digitalization, Fintech applications focus on developing and promoting innovative tools to meet customers' financial needs (Belanche *et al.*, 2019). Banks create algorithms that help them implement customer-centered strategies. Chatbots are one of the most frequently used types of

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AI. Research indicates that the use of chatbots affects individuals' acceptance of technology in all areas of human life, including banking (Sanny *et al.*, 2020).

Achieving customer satisfaction through digital platforms by offering new products and services is both a purpose and a tool for banks. However, very few studies (Trivedi, 2019) focus on customers' experience with chatbot services that banks have begun offering. This study contributes to the chatbot service literature in the banking context in two ways:

(1) investigating customer satisfaction, in line with the expectations confirmation theory (ECT); and

(2) establishing the effects of perceived trust and corporate reputation on customer satisfaction.

The rest of this study is organized into five sections. Section 2 discusses the study's theoretical background and hypotheses. Section 3 describes the study's methodology, questionnaire design, data collection method and sample. Section 4 presents data analysis and empirical findings. Finally, Section 5 discusses the results, implications, limitations and suggests further research.

Theoretical background and hypotheses development

This study relies on ECT to determine the relationships between customer expectations, perceived performance, confirmation of customer expectations and customer satisfaction. Moreover, it investigates the relationship between perceived trust, banks' reputation and customer satisfaction.

Chatbots

Chatbots represent one of the most frequently used types of human–computer interaction (HCI). They are virtual customer assistants and "AI technology products that can ask and answer customers through online channels via text message and web chat" (Lui and Lam, 2018, p. 271). A chatbot is an application to which humans may ask their questions freely and without limitation (Karri and Kumar, 2020). Chatbots may be provided with a messaging infrastructure or even voice and provide information within the time frame and limits determined by customers. Chatbots that interact with users through text or voice are frequently used in sales, marketing, customer support, technical support and training (Smutny and Schreiberova, 2020). Banks often use chatbots in customer relations management, sales, marketing activities, investment analysis and recommendations. They provide fast services and solutions, personalized digital services and cost-effective access to customer service. The operational efficiency provided by chatbots is undisputed. A Juniper Research study notes that with chatbots, banks may achieve annual cost savings of over US$8bn by 2022 (Juniper Research, 2019).

Although banks have not yet reached the targeted chatbot use, this service is offered to customers by several banks in Turkey. Aktas (2019) investigates the use of chatbots in the Turkish banking sector; the results show that 13% of the banks deliver a chatbot experience to customers through various platforms, while 9% have a developing chatbot project, and 20% include chatbot use in their two-year strategic plans. In addition, 67% of the banks using chatbots signal this service on their website. Chatbots are often employed for deposit calculation (15%), credit interest calculation (11%), social chats (8%), credit card information (7%) and exchange or stock information (4%). The Turkish banking system is progressing toward the adoption of chatbot services. Moreover, Turkey's young population, high education level and high smartphone penetration rate facilitate the spreading of chatbot services in the banking sector. The specific subject and

field of application of this study are chatbots that provide services via voice and text to bank customers in Turkey.

Expectations confirmation theory

The ECT proposed by Oliver (1980) is a customer behavior model used to define and predict customer satisfaction. It assumes that customer satisfaction is a foundation of behavior, which results from customer expectations and their confirmation or disconfirmation because of service performance (Morgeson, 2013). ECT is used to analyze pre-behavioral customer expectations and post-behavioral perceived performance in a purchase and consumption transaction (Lin *et al.*, 2005).

ECT proceeds in four stages (Oliver and De Sarbo, 1988). In the first stage, customers have expectations regarding a product or service before purchasing it. In the second stage, the purchase takes place and is affected by customers' expectations. When a substantial difference is perceived between realized performance and customer expectations, the perceived performance will increase (or decrease) with expectations. In the third stage, perceived performance either supports or rejects customers' expectations. In the fourth stage, customer expectations and perceived approval levels will affect customer satisfaction, which increases if customer expectations are verified. Hence, ECT explains customer satisfaction with customer expectations, perceived performance and the confirmation of customer expectations. This theoretical framework is frequently used to assess users' satisfactory and unsatisfactory experiences regarding a bank's information system (Bhattacherjee, 2001).

Customer expectations

Expectations may be defined as what customers believe they should and will receive in relation to the product or service or seller involved in a transaction (Kim, 2012). Customer expectations are suggestions about a product or service and describe standard features of performance (Zeithaml *et al.*, 1993). They may vary across customers and change over time. In the past, "customers expected service companies to do what they are supposed to do" (Parasuraman *et al.*, 1991, p. 40). Today, customer expectations correspond to features that individuals expect to be included in a product to make it attractive (Kosiba *et al.*, 2020). At this point, the degree of customer expectations determines an individual's perception of the realized product or service (Oliver, 2010).

Customer satisfaction is "a post–purchase attitude formed through a mental comparison of the service and product quality that a customer expected to receive from an exchange and the level of service and product quality the customer perceives from the exchange" (Kim, 2012, p. 220). A customer's satisfaction is related to their feelings. Satisfaction is built on what the service should be and depends on customer expectations (Bell and Zemke, 1987). Customer expectations are the standards used in the evaluation of customer satisfaction (Oliver, 1980). ECT qualifies customer expectations as a crucial determinant of customer satisfaction (Koppius *et al.*, 2005). In addition, customer expectations are an antecedent of satisfaction (Anderson and Sullivan, 1993). According to ECT, a significant relationship exists between customer expectations regarding a product or service and post-experience customer satisfaction (Oliver *et al.*, 1997). In line with the literature, we suggest the following hypothesis:

H1. Customer expectations regarding chatbots have a significant and positive impact on customer satisfaction resulting from chatbot use.

Perceived performance

Performance is a combination of attributes or benefits from a product or service (Tse and Wilton, 1988). It can be evaluated across two dimensions: objective performance and

perceived performance. While the former is the product or service's actual performance level, the latter includes individual evaluations (Brill *et al.*, 2019). Perceived performance relates to how customers evaluate a product or service in terms of quality and perceive its value after purchasing it (Churchill and Surprenant, 1982). In this study, perceived performance is measured as the usefulness of chatbot use and its effects on individuals' productivity and performance. Similar strategies have been adopted in past studies addressing technology (Aref and Okasha, 2019; Davis, 1989; Venkatesh and Davis, 2000). Customers' tendency to use an application is determined by their belief in whether the application will make their job easier. This belief determines the perceived usefulness of the product or service (Davis, 1989). Perceived usefulness implies that the product or service of interest will help individuals save time and effort in their daily lives, such as the internet and related tools (Zarrad and Debabi, 2012).

Customers evaluate their post-use beliefs about a product or service's performance based on their pre-use expectations (Wolverton *et al.*, 2020). Suppose the difference between customer expectations and actual performance is perceived as small; in that case, it reveals the perceived performance level in line with expectations, and thus, perceived performance increases (or decreases) with customer expectations (Anderson and Sullivan, 1993). According to ECT, a significant relationship exists between customer expectations and post-experience performance (Oliver, 2010; Spreng and Olshavsky, 1993). Customers evaluate perceived performance according to their initial expectations. Hence, in line with previous studies, we suggest the following hypothesis:

H2. Customer expectations from chatbots have a significant and positive impact on chatbots' perceived performance.

Customer satisfaction is a global judgment on how a product's or service's performance aligns with customer expectations (Anderson and Sullivan, 1993). It depends on perceived performance and is directly related to perceived performance (Mou *et al.*, 2017). Perceived usefulness, an element of perceived performance, is one of the fundamental determinants of customer satisfaction with technological products such as mobile banking (Susanto *et al.*, 2016; Yuan *et al.*, 2016). According to ECT, a product or service's perceived performance directly affects customer satisfaction (Gupta and Stewart, 1996). Moreover, if a product or service's realized performance is high, customers are more likely to be satisfied regardless of their pre-purchase expectations (LaTour and Peat, 1979). Hence, we propose the following hypothesis:

H3. The perceived performance of chatbots has a significant and positive impact on customer satisfaction resulting from chatbot use.

Confirmation of customer expectations

Confirmation of customer expectations occurs when the product or service's perceived performance coincides with customers' pre-purchase expectations or a positive performance evaluation (Kim, 2012). Confirmation of customer expectations is a cognitive process involving a comparison of customers' pre-purchase expectations and post-purchase evaluations. In the technology field, it expresses the extent to which customers' expectations for a product or service are met by their experience of use (Huang, 2019). In this study, confirmation of customer expectations implies meeting the benefits, advantages and usefulness expected by customers when using a chatbot service.

Customer expectations relate to the characteristics and qualifications of a product or service. The confirmation process, which is the customer's assessment of the performance of a product/service, is based on the matching of customer expectations pre-purchase and the performance result (Kim, 2012). Essentially, confirmation of customer expectations is the

fulfillment of customer expectations. Confirmation of customer expectations in the service sector only occurs when the service provider offers the expected benefits (Bhattacherjee, 2001). According to ECT, a positive relationship is observed between customer expectations and post-experience confirmation of customer expectations (Chou *et al.*, 2010; Lankton *et al.*, 2014). Based on these arguments, we propose the following hypothesis:

H4. Customer expectations from chatbots have a significant and positive impact on the confirmation of customer expectations from chatbot use.

ECT assumes that customer expectations and perceived performance are the premises of confirmation of customer expectations. Perceived performance is how the customers evaluate the product/service in terms of quality and perceive its value after purchasing it (Churchill and Surprenant, 1982). On the other hand, confirmation of expectations is the customers' pre-experience expectations for the product/service and their judgment regarding post-experience performance comparison (Bhattacherjee, 2001). In reality, confirmation of customer expectations is only possible with a positive evaluation of perceived performance (Kim, 2012). Recent studies on the perceived performance and confirmation of customer expectations from technology products show that the higher the perceived performance, the more likely the confirmation of expectations (Chen and Wu, 2018). The perceived performance of a product directly impacts the confirmation of customer expectations from the product, especially in technology-based services (Brill *et al.*, 2019). Based on these arguments, we propose the following hypothesis:

H5. Chatbots' perceived performance has a significant and positive impact on the confirmation of customer expectations from chatbot use.

Customer satisfaction may be defined as "a psychological or affective state related to and resulting from a cognitive appraisal of the customer expectation-performance discrepance/confirmation" (Bhattacherjee, 2001, p. 354). In addition to customer expectations, ECT questions customer satisfaction by focusing on pre-use customer expectations and post-use performance or usability comparisons (Rahi and Ghani, 2018). If the expectations of the customers before purchasing the service/product are met after the experience, this situation is an indication that the customer expectations have been confirmed positively. Positive approval occurs when the perceived performance exceeds the customer's pre-experience expectations, and thus, the verification of customer expectations positively affects customer satisfaction (Jamal and Naser, 2002; Wu, 2013). Positive confirmation of customer expectations expresses customer satisfaction. Recent studies verify the relationship between confirmation of customer expectations and customer satisfaction (Dai *et al.*, 2020; Kim, 2018). Based on these arguments, we propose the following hypothesis:

H6. Confirmation of customer expectations from the use of chatbots has a significant and positive impact on customer satisfaction with chatbot use.

Perceived trust

Trust involves concerned parties' confidence in each other's credibility in an exchange relationship (Morgan and Hunt, 1994). In other words, it is the expectation that the counterpart will not take an opportunistic approach (Rousseau *et al.*, 1998). The literature describes trust in Web-based applications or technological products as "morality" or "reliability" (Muir, 1994; Muir and Moray, 1996). Trust is examined along two dimensions: cognitive and emotional trust. The former is concerned with the awareness of what a customer knows, and the latter, with a customer's awareness about the unknown (Komiak, 2003). Cognitive trust is based on realistic evaluations that the customer's service channel will have the necessary competence, benevolence and integrity to be trusted in the future (Komiak, 2003). This research is based on cognitive trust.

The most frequently used trust dimensions in the literature are as follows (Gefen, 1997):

(1) Competence: "Competence is ability of the trustee to do what the trustor needs" (McKnight *et al.*, 2002, p. 337). The chatbots' competence dimension points to their understanding of the customer and the relative development of appropriate behavior.

(2) Integrity: Integrity implies that "trustee honesty and promise-keeping" (McKnight *et al.*, 2002, p. 337). The integrity dimension of trust in chatbots is based on their neutral and honest approach when interacting with customers.

(3) Benevolence: "Benevolence is trustee caring and motivation to act in the trustor's interests" (McKnight *et al.*, 2002, p. 337). The benevolence dimension of trust in chatbots implies that they value customers' interests and needs higher than those of other stakeholders.

Trust is crucial in the use of technological applications. Lack of trust is a substantial obstacle to satisfaction and the use of technological applications (Gefen *et al.*, 2003). Trust encourages customers to trade with a business, especially in transactions with no interaction with a salesperson (human). The trust model by Singh and Sirdeshmukh (2000) assumes that perceived trust before a certain event will increase consumer satisfaction after the purchase experience. Hence, trust is a key step in creating a satisfied customer (McKnight and Chervany, 2002). To achieve satisfactory use experiences with virtual applications that have not been experienced before, customers must first trust the product or the business (Kim, 2012). In technology-based applications, where customers' experience is scant, trust effectively reduces customer uncertainty and risk concerns about the product (McKnight *et al.*, 2002). Hence, trust has a significant impact on customer satisfaction. The higher the trust, the higher the customer satisfaction (Morgan and Hunt, 1994). Past research has also confirmed the positive effect of trust on customer satisfaction (Akter *et al.*, 2013; Chiu *et al.*, 2009; Mulia *et al.*, 2020). Based on the literature, the following hypothesis is proposed:

H7. Perceived trust in a bank's chatbot service has a significant and positive impact on customer satisfaction from chatbot use.

Corporate reputation

Corporate reputation is a collective representation of a business's past actions and results, which describe its ability to deliver to multiple stakeholders (Fombrun *et al.*, 2000). Corporate reputation is also accepted as an external reflection of the business value (Chun, 2005). It is defined as a "customer's overall evaluation of a firm based on his or her reactions to the firm's goods, services, communication activities, interactions with the firm and/or its representatives or constituencies (such as employees, management, or other customers) and/or known corporate activities" (Walsh and Beatty, 2007, p. 129). A business' reputation is essential for reducing customers' uncertainty and anxiety about their products or services. Hence, reputation may play a crucial role in reducing the risks and uncertainties perceived by customers in the banking sector (Cintamür and Yüksel, 2018).

When customers have little information about a business's products or services, they rely on the business's corporate reputation to assess the quality of the product or service and predict their future attitudes toward it (Nguyen and Leblanc, 2001). Hence, corporate reputation influences customers' attitudes and satisfaction (Yoon *et al.*, 1993). Empirical evidence shows that corporate reputation has a significant impact on customer satisfaction. The existence of a positive relationship between reputation and satisfaction may be explained by the cognitive dissonance theory of Festinger (1957). According to this theory, a company's poor public reputation may encourage customers to reconsider their perceptions of the company's products and services, thus affecting customers' perceptions of their

experiences with such products or services. While a weak or negative corporate reputation negatively affects customer satisfaction, a positive or strong reputation positively impacts customer satisfaction regarding a product or service. Previous studies find a positive relationship between corporate reputation and customer satisfaction (Davies *et al.*, 2004; Helm, 2006). Hence, we formulate the following hypothesis:

H8. A bank's corporate reputation has a significant and positive impact on customer satisfaction from chatbot use.

Figure 1 describes the proposed theoretical framework and the related research hypotheses.

Research methodology

Sample and data collection

A questionnaire was designed to investigate the proposed research hypotheses. The questionnaire was shared with approximately 700 people via social media and message applications (Instagram, LinkedIn and WhatsApp). Participants were requested to share the questionnaire with their network. Thus, convenience sampling and snowball sampling methods were used together.

The study sample consists of individuals over the age of 18 that experienced the chatbot services offered by various banks in Turkey. The survey was completed between October 2019 and December 2019. However, as banks have newly implemented chatbots, only 256 bank customers who had experienced a bank's chatbot service could be identified. Among these, 16 provided incomplete responses. Hence, the final sample used in the data analysis comprises 240 cases.

The demographic characteristics of the study's participants are shown in Table 1. Overall, 47% of the participants are women (113), 65% (157) of the participants are married, 63% (150) are between 26 and 40 years old, 62% (148) hold undergraduate degrees and 41% (98) declared a monthly income of US$0–US$1,000.

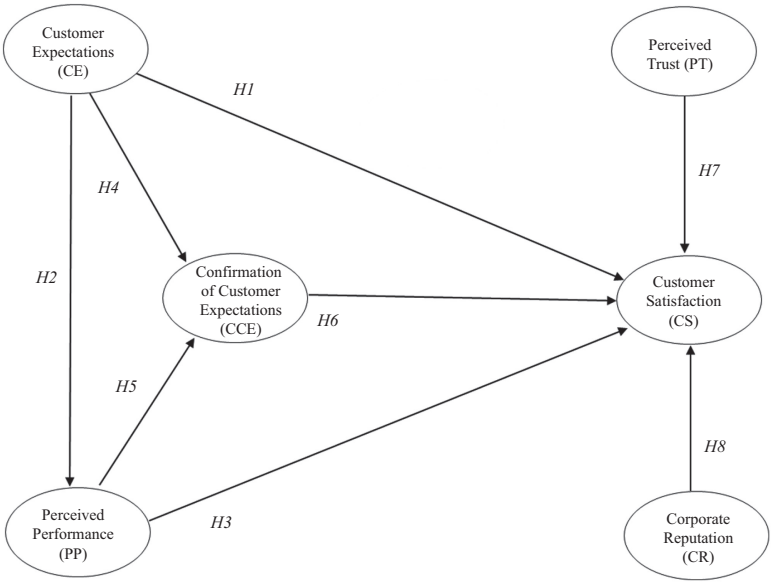


Figure 1.
Research model

Measures

Before starting the survey, a brief description of chatbots was provided. We requested participants their evaluations of the bank's services offered through chatbots (operating via text and voice). In the first part of the survey, we investigated customer expectations from chatbots and their experience of using them, while in the second part, we focused on the demographic characteristics of the participants.

All scale items used in the study were adapted from the literature. Customer expectations, perceived performance and confirmation of customer expectation scales were obtained from Davis (1989), the perceived trust scale from Xiao and Benbastat (2002), the corporate reputation scales from Özkan *et al.* (2019) and the customer satisfaction scale from Chung *et al.* (2018). Multiple items were used to measure all constructs. The customer expectations, perceived performance and confirmation of customer expectations scales include six items: the corporate reputation scale comprises four items, and the customer satisfaction scale includes six items. The perceived trust scale was assessed using three subtitles and 12 items (competence – five items, benevolence – four items and integrity – three items). Moreover, all items were evaluated using a five-point Likert-type scale ranging from "Strongly disagree" (1) to "Strongly agree" (5) or "Much worse than expected " (1) to "Much better than expected " (5). A questionnaire comprising these items was pre-tested. In addition, it was thoroughly reviewed and finalized by academic and banking sector experts.

Results

The survey data were analyzed using explanatory factor analysis (EFA). Each expression was loaded on its own factors, with factor loadings above 0.60. Also, the correlation between the sub-dimensions was examined. A high correlation value indicates contiguity, while a low value implies segregation. The expected result is neither very high nor low. A value above 0.70 implies a high-level relationship between the variables (Tabachnick and Fidell, 2013). Hence, a total of nine items (2 CE, 2 CCE, 1 PP, 2 CS, 1 PT and 1 CR item) above the correlation value of 0.70 were excluded from the analysis. Table 2 reports satisfactory results for both independent and dependent variables, as well as the means and standard deviations measures.

To test whether the measurement model meets the required conditions, validity and reliability analyses were performed on the latent variables using confirmatory factor analysis (CFA), and the measurement ability of the model was tested. The structural equation model

Gender			Marital status			Age		
Description	Count	%	Description	Count	%	Description	Count	%
Female	113	47	Married	157	65	18–25	20	8
Male	127	53	Single	83	35	26–40	150	63
						41–62	63	26
						Over 62	7	3

Education			Monthly income		
Description	Count	%	Description	Count	%
Primary	0	0	0–1000 $	98	41
High school	15	6	1000–2000 $	80	33
Junior college	23	10	2000–4000 $	45	19
Undergraduate	148	62	Over 4.000 $	17	7
Master's/PhD	54	23			

Table 1. Sample characteristics (n = 240)

(SEM) was employed to test the relationship between the variables and the proposed hypotheses. All analyses were performed using SPSS 21 and SmartPLS programs. Smart PLS-SEM analysis was preferred, as it is particularly suitable for small sample sizes.

Confirmatory factor analysis

A measurement model consisting of six latent variables and 31 indicators was evaluated using CFA. As a first step, the fit values of the model were examined. The compatibility indexes (NFI = 0.902, SRMR = 0.045) related to the model were obtained. The results indicate that the model seems to fit the data well (Kline, 2005).

The internal consistency (Cronbach's alpha) value is used as a criterion for transforming first-level variables into a higher-level factor structure (Brakus *et al.*, 2009). In research, Cronbach's alpha has been used to separately assess the reliability of measurement scales. The results in Table 2 indicate that the reliability of the scales according to Cronbach's alpha values ranges between 0.811 and 0.883. As the Cronbach's alpha values for all scales are greater than the guideline of 0.70, the scales exhibit acceptable reliability (Saunders *et al.*, 2003). However, more than one heterogeneous but similar expression, such as the composite reliability (CR) value used to measure the overall model's reliability, contributes to ensuring that a scale is reliable. The values for the scales are listed in Table 2. The reliability of the scales is high (>0.70), indicating that construct validity is adequate (Fornell and Larcker, 1981).

Content validity is acceptable, as the proposed scale items have been widely used in the literature and expert evaluation. However, convergent validity was examined. Three conditions are required to ensure convergent validity. The standard factor loading of each observed variable belonging to latent variables should be greater than 0.50, the reliability for each sub-dimension should be greater than 0.70 and statistically significant and each mean residual variance (AVE) should be at least 0.50 (Fornell and Larcker, 1981). Table 2 shows that the CR of each scale is over 0.90, the Cronbach's alpha reliability values are higher than 0.80 for each sub-dimension and all sub-dimensions have high reliability. The AVE value for each sub-dimension is greater than 0.50. In addition, the factor loadings of the items in each dimension are higher than the minimum and maximum values of the factor loadings and statistically significant. Therefore, convergent validity is supported.

The relationship between the variables of interest was examined using the Pearson correlation coefficient. This parametric method is used to explore the relationship between continuous variables, when the sample size is sufficiently large ($n > 30$). The relationship between all variables is statistically significant at the 1% level ($p < 0.01$), and all correlation values are positive. One of the crucial assumptions in SEM and regression analysis is that a significant relationship exists between the dependent and independent variables. In this study, the relationship between customer satisfaction and other variables is significant. The results in Table 3 show that perceived trust has the most substantial relationship with

Table 2. Summary of the results for reflective measurements

Variable	Item	Factor loads	Mean	SD	AVE	CR	KMO	Cronbach's alpha
CE	4	0.789–0.897	2.736	1.465	0.738	0.918	0.780	0.883
PP	5	0.844–0.923	3.106	1.252	0.782	0.947	0.869	0.820
CCE	4	0.900–0.923	3.136	1.105	0.834	0.953	0.867	0.824
PT	11	0.738–0.850	3.039	1.210	0.637	0.951	0.941	0.823
CR	3	0.917–0.944	3.057	1.271	0.963	0.950	0.749	0.811
CS	4	0.871–0.923	3.157	1.213	0.821	0.948	0.851	0.817

customer satisfaction, as the correlation coefficient between the two is $r = 0.688$. In other words, as perceived trust increases, customer satisfaction increases substantially.

Testing the proposed hypotheses: structural equation model

Partial least squares (SmartPLS) SEM was used to test the research hypotheses. When the assumptions required in multivariate statistics (normal distribution) are not met, PLS-SEM is recommended, as it is considered a flexible method. When the sample is large enough, this approach yields robust results despite missing data (Hair *et al.*, 1998).

Figure 2 shows that customer expectations have no significant impact on customer satisfaction ($\beta = -0.085$, $p = 0.061$); hence, H1 is not supported. Customer expectations have a significant impact on perceived performance ($\beta = 0.377$, $p = 0.000$); hence, H2 is supported. Similarly, perceived performance has a significant impact on customer satisfaction ($\beta = 0.203$, $p = 0.016$), supporting H3. While customer expectations have a positive effect on perceived performance, they have no impact on customer satisfaction. However, perceived performance has a positive effect on customer satisfaction. Hence, customer expectations may have an indirect impact on customer satisfaction through perceived performance. Moreover, the impact of customer expectations on the confirmation of customer expectations is not significant ($\beta = 0.081$, $p = 0.260$); hence, H4 is not supported. However, the impact of perceived performance on the confirmation of customer expectations is statistically significant ($\beta = 0.284$, $p = 0.001$), supporting H5. Confirmation of customer expectations has no significant impact on customer satisfaction ($\beta = 0.041$, $p = 0.461$); therefore, H6 is not supported. Finally, Figure 2 shows a significant impact of perceived trust on customer satisfaction ($\beta = 0.389$, $p = 0.000$) and a significant impact of corporate reputation on customer satisfaction ($\beta = 0.354$, $p = 0.000$). Therefore, H7 and H8 are supported. All the proposed hypotheses, except H1, H4 and H6, are supported (Table 4).

The R^2 values are 0.104, 0.142 and 0.617 for the confirmation of customer expectations, perceived performance and customer satisfaction for chatbot use, respectively. In other words, 10.4% of the confirmation of customer expectations, 14.2% of the perceived performance and 61.7% of customer satisfaction is explained by the related variables.

Discussion and conclusion

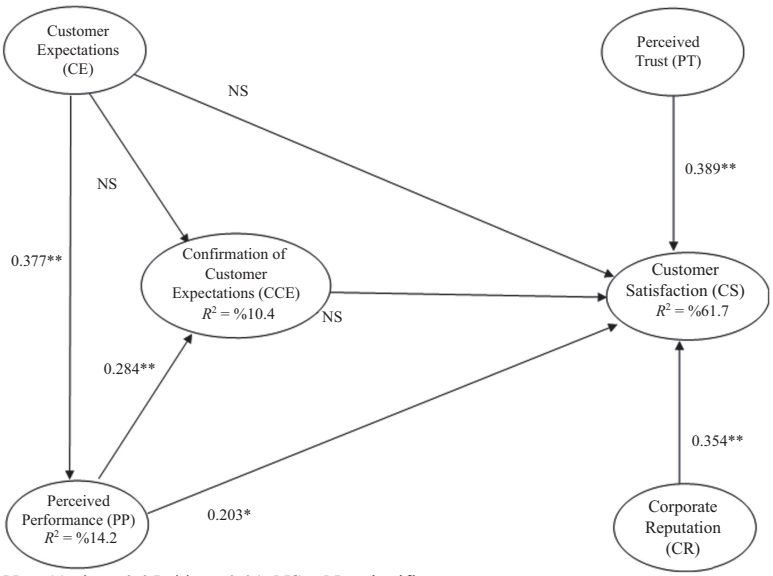
The framework of the study is based on the theory of confirmation of customer expectations. In addition to ECT, the relationships between perceived trust, corporate reputation and customer satisfaction were also examined.

We hypothesized that the primary determinants of customer satisfaction are customer expectations, perceived performance, confirmation of customer expectations, perceived trust and corporate reputation. The results confirm that performance perceptions regarding the use of chatbot positively affect customer satisfaction. After the chatbot service is

	CE	CCE	PP	CS	CR	PT
CE	*0.738*					
CCE	0.189**	*0.834*				
PP	0.365**	0.309**	*0.782*			
CS	0.235**	0.336**	0.653**	*0.821*		
CR	0.319**	0.338**	0.510**	0.624**	*0.963*	
PT	0.336**	0.337**	0.641**	0.688**	0.473**	*0.637*

Note(s): The values in *italics* represent the AVE statistic, $^{**}p < 0.01$

Table 3. Correlation between the variables



Note(s): $*p < 0.05$, $**p < 0.01$, NS = Not significant

Figure 2.
The standardized path coefficients and significance levels

experienced, the benefits in terms of productivity and efficiency play a crucial role in customer satisfaction. Unlike internet banking and mobile banking, personalized services through chatbots provide substantial convenience and benefits to customers. This result is consistent with Lin *et al.* (2009), Tsai *et al.* (2014), Yuan *et al.* (2016), Susanta *et al.* (2013) and Shiau *et al.* (2020). Similarly, customers' perceived trust in the bank's chatbot service positively affects customer satisfaction. Consumers have initial concerns regarding technology-based financial products or services. Such worry and insecurity hinder satisfaction, but trust in chatbot services positively affects customer satisfaction. This result is parallel to Akter *et al.* (2013), Mulia *et al.* (2020), Ofori *et al.* (2018) and Sarkar *et al.* (2020). Also, banks' reputation positively impacts customer satisfaction. A business's reputation may generate customer satisfaction when customers cannot fully control the scope and process of transactions, especially in banking. Customers' experience with chatbots is

Table 4.
Outcome of the path analysis

Hypothesis	Parameter	*t*-value	*p*	Result
H1. Customer expectations → Customer satisfaction	−0.085	1.876	0.061	Not supported
H2. Customer expectations → Perceived performance	0.377	6.239	0.000	Supported
H3. Perceived performance → Customer satisfaction	0.203	2.410	0.016	Supported
H4. Customer expectations → Confirmation of customer expectations	0.081	1.128	0.260	Not supported
H5. Perceived performance → Confirmation of customer expectations	0.284	3.438	0.001	Supported
H6. Confirmation of customer expectations → Customer satisfaction	0.041	737	0.461	Not supported
H7. Perceived trust → Customer satisfaction	0.389	4.335	0.000	Supported
H8. Corporate reputation → Customer satisfaction	0.354	4.855	0.000	Supported

limited. Hence, the positive reputation of a bank offering the chatbot service has a positive effect on customer satisfaction. This result is supported by Loureino and Kastenholz (2011), Su *et al.* (2016) and García-Madariaga and Rodríguez-Rivera (2017).

The study's results indicate that customer expectations for chatbot services have no impact on customer satisfaction. This result contradicts the literature. Individuals have clear expectations for products that they experienced and know well. Expectations correspond to the standards that a service or product should have. Moreover, customer expectations may be affected by the experiences of other customers, and this can shape satisfaction differently. As chatbots are in the introduction stage of the product/service life cycle and very few people experienced chatbots in Turkey, in contrast with internet banking or mobile banking, this situation may have eliminated the predicted effect of customer expectations on customer satisfaction. Although customer expectations regarding the chatbot service do not affect customer satisfaction, they influence perceived performance. In turn, perceived performance affects customer satisfaction, particularly in the case of positively perceived performance. This result suggests that customer expectations may have an indirect impact on customer satisfaction through perceived performance. The confirmation of customer expectations regarding chatbots does not affect customer satisfaction. Although customers' perceptions about the performance of the chatbot service positively affect customer satisfaction, the uncertainty in customer expectations affects the results. In addition, although the use of chatbot provides convenience and creates various benefits in customers' lives, the general concerns or privacy worries of the participants regarding the product, bank, technological infrastructure and security structure of the bank may hinder customer satisfaction. This result contradicts Lin *et al.* (2009), Tsai *et al.* (2014) and Rahi and Gani (2018).

The study's results show that customer expectations regarding chatbots positively affect their perceived performance. Customers perceive the difference between chatbot service expectations and actual performance as low or zero. Hence, perceived performance depends on customer expectations. This result is supported by Lankton *et al.* (2014). In addition, customers' perception of chatbot performance has a positive impact on the confirmation of customers' expectations. It shows that customers' expectations are confirmed only when perceived performance is high. This result is in line with Lin *et al.* (2005) and Kim (2012). However, customer expectations regarding chatbots do not affect the confirmation of customer expectations. Chatbots are a novel banking service; hence, customer expectations for chatbot services may differ, especially regarding their usefulness. In line with previous results, this study verifies that the confirmation of customer expectations regarding chatbot services is possible when perceived performance is high. The heterogeneity of the participants in terms of demographic characteristics and technological competence might have affected this result. This conclusion is supported by previous research (Venkatesh *et al.*, 2003; Lankton *et al.*, 2014).

Today, innovation spreads very fast, especially in technological service channels. Providing chatbots to their customers as an active service delivery channel may favor banks in terms of competition. Banks should develop trust in their technological products, increase their chatbot service's performance beyond customer expectations and further improve their corporate reputation to achieve customer satisfaction. As the chatbot service is in the introductory phase, customer experience should be questioned more frequently. Thus, banks should conduct customer interviews and learn about customers' expectations, especially through open-ended questions, to review their performance and technological investment. Positive experiences with chatbots may induce customers to encourage people around them to use this service channel, increasing the rate and frequency of chatbot use. As a result, customers may use their time more effectively and receive 24/7 personal banking services. Banks may decrease their transaction costs and increase efficiency. Nowadays, individuals often refrain from face-to-face

communication in their relations; hence, they may favor chatbots in their banking transactions. The importance of chatbots in providing personal services is particularly evident when personal contacts are limited due to events such as the recent COVID-19 outbreak.

Research's implications for practice and society

Chatbots deliver information in the form of questions and answers through voice or text messages, and they offer dialogue-based communication. For banks, chatbots are an effective method for managing sales and customer services because of their fast and effective communication features. Banks may improve customer satisfaction resulting from chatbot use by enhancing their product and service performance, their reputation and trust in terms of technology use and credibility. Especially in extreme events, such as the COVID-19 outbreak, chatbots are crucial service delivery channels, which ensure the reception of personal services when banks work through minimum individual communication and maximum technology usage. Although chatbots initially require relatively high investment in technology, they can lower transaction costs and personnel costs and offer higher profit and working volumes with improved customer experience. As individuals' adaptation and access to technology increase, chatbots are expected to increase their service delivery share in the banking sector.

Limitations and future research

This study has some limitations. First, owing to the convenience sampling technique, the sample is not representative of the entire target population, and the results should not be generalized to the whole population of Turkey. Second, this research reflects the results for an emerging market country with relatively high technology sensitivity and a young population. Third, chatbots are at the beginning of their product life cycle, and the number of customers who experienced the chatbot service is limited. Therefore, this study was conducted using a limited number of participants. As the number of bank customers experiencing chatbots increases, future research may require adopting different perspectives. The impact of different variables should be explored. Customers' demographic characteristics and technology readiness should be investigated, as well as customers' satisfaction with chatbots, by addressing financial products and confidentiality issues. In addition, the impact of the development of chatbots' empathy, inquiry skills and gender characteristics on customer satisfaction should be assessed in future studies. Moreover, the adoption of and customer satisfaction levels for chatbot services may differ across countries depending on their level of development. Therefore, future studies should address developed countries to confirm the relationships tested in the present study. Finally, this research focused on customer satisfaction; the intentions and other attitudes of prospective customers have not been questioned. Future research should investigate loyalty and word-of-mouth (WOM) communication in relation to the effects assessed in this research and the topics suggested above.

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About the author

Berrin Arzu Eren is an Assistant Professor at Ufuk University. She holds a PhD degree in Anadolu University Marketing Science. Her fields of work are service marketing, financial marketing, customer relationship management and sales management. She had 13 years of banking experience, and she managed Garanti Bank BBVA's several branches in Ankara and Istanbul. Berrin Arzu Eren can be contacted at: berrinarzu.eren@ufuk.edu.tr

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